

Hammer boundary

$$J_h, \theta_h, \omega_h$$

$$\tau_c \omega_h$$

Contact

$$\delta = \theta_h - \theta_a$$

$$\tau_c \omega_a$$

Anvil

$$J_a, \omega_a$$

$$\tau_j \omega_a$$

Socket and joint

$$k_j, c_j$$

Elastic store U_c
and dissipation D_c

Elastic store U_j
and dissipation D_j

Unknown component

$$\tau_X = B_3 \theta_h^{(3)} + B_4 \theta_h^{(4)}$$

$$P_X = \tau_X \omega_h$$